SentryLoop: Tile QC Inspector

Technical report · Team Cloud Catalyst (Edmund Anthony, Nandani Chauhan) · OpenCV AI Competition 2026, Agentic Vision · October 2026

1.00 / 0.95	**0**	**~60 ms**	**100%**
precision / recall, unseen set	false alarms on 40 good tiles	per tile, laptop CPU	of rejects wait for a human

1. Problem

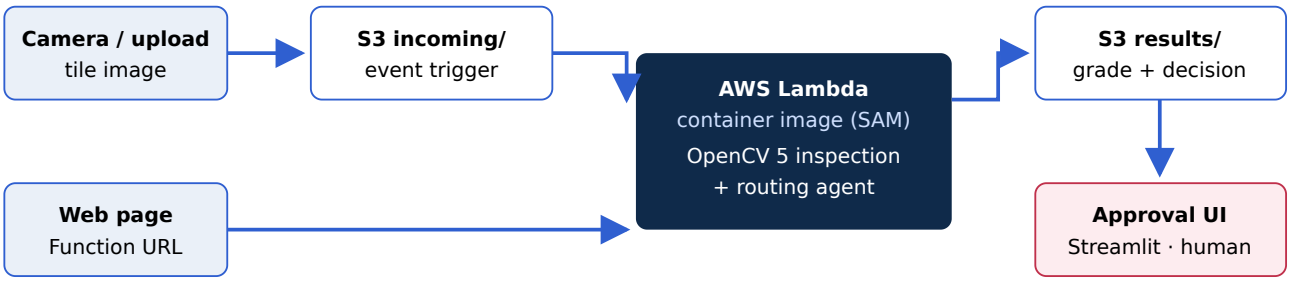
At the end of a ceramic tile kiln, people still check every tile by eye for cracks, chips, spots and stains, then decide whether it is first quality, second quality, or scrap to be crushed and recycled. The work is slow and subjective, and slate- or stone-effect tiles make it harder because no two tiles share the same surface texture.

SentryLoop automates the inspection and the routing decision, but keeps a person in charge of the one costly action: sending a tile to recycling.

2. Solution overview

1. **Learn "good":** about 20 of the best tiles are run past the camera; SentryLoop learns the normal range of their surface statistics.
2. **Inspect:** each new tile is located, straightened and checked for cracks, chips, spots and stains with OpenCV 5.
3. **Grade:** A, B or REJECT from defect type and size.
4. **Act:** an agent routes A → Packer 1 and B → Packer 2 automatically; REJECT → Recycle only after a supervisor approves (or overrides to B).
5. **Log:** every result, decision and approval is stored in Amazon S3.

3. Architecture on AWS



The same Lambda serves both paths; approvals are written back to S

- **Amazon S3:** `incoming/` (uploads, triggers Lambda), `results/` (inspection + decision JSON), `approvals/` (human decisions). A 14-day lifecycle rule keeps storage cost near zero.
- **AWS Lambda (container image):** OpenCV is too large for a zip package, so the function ships as a Docker image built and deployed with AWS SAM (Python 3.12, OpenCV 5.0.0, 2 GB memory). A Function URL exposes a public test page.
- **Approval UI (Streamlit):** reads `results/`, draws the defect boxes on the source image, and records *Approve recycle* or *Override to B*.

- **Optional:** Amazon Bedrock can write a plain-English end-of-shift note from the decisions; it never makes a decision.

4. Vision pipeline (OpenCV 5)

Step	Method
Locate and straighten	Gaussian blur, Otsu threshold, morphological close, largest contour, minimum-area rectangle, perspective warp to 1200 × 600 px
Remove lighting	High-pass: image minus a large Gaussian blur (σ = 25), so gradients from the light source disappear
Defect maps	Dark and bright deviation maps (cracks, pits, spots); an edge band check for missing material (chips); 50 px patch statistics for stains and shade
Learned thresholds	Extreme percentiles of the reference tiles' own maps, so "normal texture" is learned per tile design, not hand-tuned
Classify	Connected components; shape rules (length, and how much of its bounding box a blob fills) separate thin cracks from compact spots; edge holes become chips

> **Why statistics, not pixel matching:** slate- and stone-effect tiles have a different random texture on every piece, so comparing against one "golden image" flags the texture itself. Learning the normal range from about 20 good tiles makes anything outside it stand out.

Grading and routing rules (deterministic and explainable)

Condition	Grade	Action	Human gate
Any crack or chip	REJECT	Divert to recycle	Yes: waits for approval
Any stain, more than 3 defects, or strong shade deviation	B	Packer 2 (second quality)	No
Spots larger than 120 px	B	Packer 2	No
Tiny spots only, or no defects	A	Packer 1 (first quality)	No

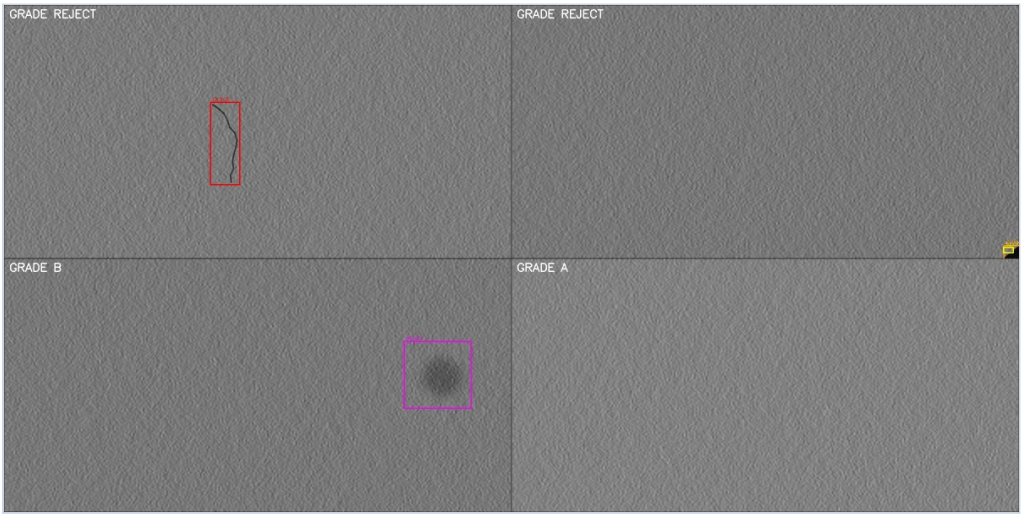


Figure 1. Overlays produced by the pipeline: detected defects are boxed and labelled with the resulting grade.

5. Evaluation

Public data is synthetic: a generator creates slate-effect tiles (20 reference tiles, plus test sets of 40 tiles each: 20 defect-free and 5 each of crack, chip, spot and stain). The reference was learned once from the demo set's 20 reference tiles and then applied unchanged to an unseen set generated with a different seed.

Set	Tiles	TP	FP	FN	TN	Precision	Recall
Demo (seed 7)	40	20	0	0	20	1.00	1.00
Unseen (seed 99)	40	19	0	1	20	1.00	0.95

Set	Exact grade agreement	Misses
Demo	39 / 40	1 small spot graded A instead of B (tiny spots are allowed in A by design)
Unseen	37 / 40	2 small spots graded A; 1 small edge chip graded A instead of REJECT

Speed: about 60 ms per tile for inspection on a laptop CPU (1200 × 600 px). **Live check on AWS (25 Oct 2026):** tile_011 → A, Packer 1, auto-executed; tile_019 → REJECT (crack), awaiting approval; tile_037 → REJECT (chip, spot), awaiting approval; the function reports OpenCV 5.0.0.

Reproduce: `python pipeline/make_demo_tiles.py`, `python pipeline/tile_qc.py build --refs samples/demo/reference`, `python pipeline/tile_qc.py eval --ref models/reference.json --test samples/demo/test`.

6. Agentic behaviour and human oversight

- The vision result changes what the line does next: each tile is routed, not just labelled.
- Low-risk actions (A and B routing) run automatically; the one irreversible, costly action (recycling) always waits for a person.
- The rules are deterministic, so every decision can be explained from the defects found. The missed edge chip on the unseen set is the reason rejects keep a human gate and why a lower chip threshold is a deliberate business trade-off (fewer misses, more false rejects).

7. Limitations

- Public results are on synthetic tiles, which are easier than real line images (even lighting, one tile per frame, known defect sizes).
- The reference is learned for one tile design; each new design or glaze needs its own reference set.
- Very small edge chips can be missed (1 of 40 on the unseen set).
- Real-camera validation so far is a small private sanity check; no factory images are published.
- Lambda cold start adds a few seconds to the first request after idle.

8. Responsible use

No factory or personal images are in the repository; real images stay local and git-ignored. AI supports, never replaces, the supervisor's decision on scrap. No credentials are stored in the

code; the deployment uses scoped IAM permissions to the stack's own bucket.

9. Next steps

- Validate on real line cameras, including multi-light (grazing-light) captures.
- Colour-shade grading per production batch.
- A per-shift dashboard with defect trends that point maintenance to the press or kiln causing them.

Repository: github.com/nandini9690/OpenCV-AI · Live test page: hkdbd4v3q4mfndvdjo4kvxzndm0ddjlj.lambda-url.us-east-1.on.aws